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Ultrasonics 41 (2003) 501–507
Ultrasonics
www.elsevier.com/locate/ultras

Surface impedance matrices to model the propagation in multilayered media

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Received 21 May 2003; accepted 23 June 2003

Abstract

The surface impedance matrices in stratified plates made of fluid layers and/or anisotropic absorbing solid layers link the particle velocity field to the stress field at any interface. A surface impedance matrix represents the impedance at a given interface of all the layers located between that interface and one boundary of the medium. For each interface, there are two surface impedance matrices, each one corresponding to one boundary. This notion simplifies the computations of the modal solutions. The number of elements in the matrices involved in the computations is divided by a factor of four in comparison to usual matrix methods. This paper describes the method and presents examples to illustrate its interests and its efficiency where other techniques fail, for instance in the case of modes possessing energy in layers embedded in the structure.
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Keywords: Iterative method; Stratified media; Surfaces impedances

1. Introduction

Multilayered materials can be encountered in various branches of the physics with very different scale values. Two extreme cases can be represented by seismological applications and NDE/NDT ultrasonic applications. The standard model that was introduced by Thomson [1] and Haskell [2], uses the transfer matrix method that connects state vectors from an interface to another with simple matrix multiplications. Lowe [3] wrote a recent review of the numerous papers related to matrix techniques used for modeling wave propagation in anisotropic, absorbing materials. A discussion on the stability of the numerical computations is made. There are two main approaches: the "classical" transfer matrix method that is fast but unstable when the frequency, the thickness or the number of layers becomes large, and the global method that is stable but slower since the size of the matrices is growing with the number of layers [3].

In both methods, the state vector represents displacements and stresses at the same interface. The resulting matrix that connects a state vector from an interface to another, is a combination of terms without dimension and terms equivalent to stiffness or compliance. Recently, Wang and Rokhlin [4–8] introduced a method in which the transfer matrix is a layer stiffness matrix that links a vector composed of stresses at both sides of the layer to a vector composed of displacements at the same interfaces. The main advantage of this method is that it is stable, while being as fast as the "Thomson/Haskell" method [4–8].

Another previous approach was proposed in 1972 by Li and Yen [9] and used later by Honein et al. [10]. By analogy with the electric concept, they introduced the surface impedance matrix for a semi-space constituted of many layers. The surface impedance matrix connects the stress field to the particle velocity field at a given interface, and takes into account the contribution of all the layers located between that interface and one boundary of the stratified medium. At any interface of a stratified medium, two surface impedance matrices can then be computed. They represent the impedance of the layered media located at either side of the interface.

In this paper, a general iterative scheme of the surface impedance matrix method is proposed, and combined with the formalism of the layer stiffness matrix method [4–8]. This latest is actually slightly modified so that layer impedances are calculated instead of layer stiffnesses. In this context, the layer stiffness matrix method

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doi:10.1016/S0041-624X(03)00167-7

will be called layer impedance matrix method all through the paper. The combination of the two methods reduces the number of operations at each step of the iterative process, thus leading to significant increases in the speed of computation. In comparison with usual methods [1–8], the dimensions of the matrices in the surface impedance method are divided by two. Hence the number of operations for finding modal solutions of multilayered plates located between two semi-infinite half spaces is strongly reduced. The main difference with usual methods is the fact that the impedances of the semi-infinite half spaces are immediately taken into account, i.e. before assembling the whole-medium matrix. In the usual methods, the global matrix of the layer stacking is built first, and then the contribution of the semi-infinite half spaces is introduced. Introducing this information from the beginning implies the size of the matrices and the number of computations to be reduced. Since these methods are often used for solving inverse problems, the computation time is an important issue.

The surface impedance matrix method also permits numerical instabilities associated with modes propagating along embedded layers, called "channel" modes, to be eliminated. For instance, the through-thickness stress and displacement fields of such modes tend towards zero, at the surfaces of the stratified medium. Consequently, using the usual methods, which consist in affecting a non-null value at one component of these fields, at one of the two surfaces of the medium, lead to very high values of the fields as the position through the medium gets close to the embedded layer. In these situations, these methods become numerically unstable. The surface impedance matrix method being based on the selection of a specific interface in the medium, the through-thickness fields are calculated starting from that interface, thus leading to stable and correct results.

The structure of the paper is the following: the problem of the propagation in layered media, the surface impedance matrix method and the layer impedance matrix method (modified layer stiffness matrix method) are briefly reminded. Then, an iterative scheme is proposed for the surface impedance matrix method, as well as its combination with the layer impedance matrix method. The ways to compute modal solutions and through-thickness stress or particle velocity fields are then presented. One application in relation with closed-form solutions that can be found in the literature is given to show the numerical stability of the proposed method. Finally, another application is presented for emphasizing its ability to find channel modes where closed form solutions cannot be established or where usual matrix methods fail.

2. Problem definition. Surface impedance matrices

A layered plate is included between two semi-infinite media. The problem is to compute the stress and particle velocity fields anywhere in the medium when no wave is incident (modal solutions) or when there is one incident wave from any of the two semi-infinite media.

Fig. 1 presents the geometry of the problem. The stratified medium is composed of n plane layers, made of viscous fluids or viscoelastic anisotropic solids. The interfaces are normal to the direction 1. For anisotropic solid in a non-symmetry plane (3D), the quantities that need to be evaluated at the interface j are the three

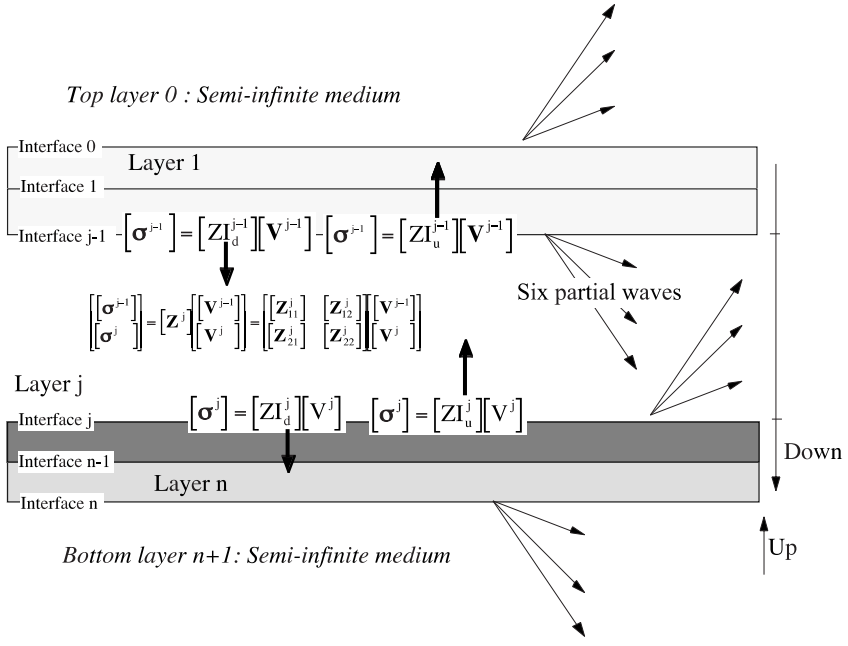


Fig. 1. Interface and layer impedance matrices in a stratified medium.

components v_1^j, v_2^j, v_3^j of the particle velocity vector and the three components σ_1^j, σ_6^j, σ_5^j of the stress tensor. In these expressions, the usual contracted notation is used [11]. For isotropic or anisotropic solids if the plane of propagation coincides with a plane of symmetry (2D) and only modes that produce displacements in this plane are considered, then only the two pairs of components v_1^j, v_2^j and σ_1^j, σ_6^j have to be computed. These are reduced to v_1^j and σ_1^j for a fluid layer. In the following, these quantities are represented by the vectors $[\mathbf{V}^j] = \begin{bmatrix} v_1^j \\ v_2^j \\ v_3^j \end{bmatrix}$ and $[\boldsymbol{\sigma}^j] = \begin{bmatrix} \sigma_1^j \\ \sigma_6^j \\ \sigma_5^j \end{bmatrix}$.

By definition [9,10], a $[ZI^j]$ surface impedance matrix connects the stress vector to the particle velocity vector at any interface j, such as:

$$[\boldsymbol{\sigma}^j] = [ZI^j][V^j]. \tag{1}$$

This notation is equivalent to the classical wave impedance [11, p. 246] with the opposed sign. The number of elements of a surface impedance matrix is 3×3, 2×2 or 1×1, depending on the nature of the layer.

The general idea of the computation is to start from interface n, where $[ZI_d^n]$ is the surface impedance matrix of the bottom semi-infinite medium and to compute "Going up" the next surface impedance matrices until the last one $[ZI_d^0]$. The same computation can be done "Going down" for all the surface impedance matrices $[ZI_u^j]$ starting from $[ZI_u^0]$ that is the surface impedance matrix of the top semi-infinite medium to the last one $[ZI_u^n]$.

When "Going up", $[ZI_d^j]$ represents the impedance of the whole medium located below the interface j ("Looking down") and when "Going down", $[ZI_u^j]$ represents the impedance of the whole medium located above interface j ("Looking up"). If the medium has no mirror symmetry with respect to the interface j, then the two matrices are different, since the impedance is different "Looking down" or "Looking up". To compute by iteration all the surface impedance matrices, a link between the two interfaces limiting the same layer must be used. This link can be either the Thomson-Haskell matrix [1,2] or the layer impedance matrix. The latest is chosen for its stability [4–8] and therefore briefly reviewed in the next section.

3. Layer impedance matrices

The layer impedance matrix computation is fully similar to the layer stiffness matrix computation presented recently by Wang and Rokhlin [4–8]. The small difference is the fact that the displacements are replaced by velocities, so that the matrix is an impedance matrix. It is briefly described here for introducing the notations.

The layer j is limited by the top interface $j-1$ and the bottom interface j (Fig. 1). The stresses $[\boldsymbol{\sigma}^{j-1}]$ and $[\boldsymbol{\sigma}^j]$ are linked to the particle velocities $[V^{j-1}]$ and $[V^j]$ at interfaces $j-1$ and j, respectively, by a matrix $[Z^j]$ called "layer impedance matrix":

$$\begin{bmatrix} [\boldsymbol{\sigma}^{j-1}] \\ [\boldsymbol{\sigma}^j] \end{bmatrix} = [\mathbf{Z}^j] \begin{bmatrix} [\mathbf{V}^{j-1}] \\ [\mathbf{V}^j] \end{bmatrix} = \begin{bmatrix} [\mathbf{Z}_{11}^j] & [\mathbf{Z}_{12}^j] \\ [\mathbf{Z}_{21}^j] & [\mathbf{Z}_{22}^j] \end{bmatrix} \begin{bmatrix} [\mathbf{V}^{j-1}] \\ [\mathbf{V}^j] \end{bmatrix}. \tag{2}$$

According to the nature of the layer, the number of elements in a layer impedance matrix (6×6, 4×4 or 2×2) is four times bigger than that in a surface impedance matrix. This matrix $[\mathbf{Z}^j]$ can be decomposed in four sub-matrices. $[\mathbf{Z}_{11}^j]$ is the layer impedance sub-matrix at interface $j-1$ in the layer j and $[\mathbf{Z}_{22}^j]$ the layer impedance sub-matrix at interface j. The two other sub-matrices are coupling the two interfaces together. All the sub-matrices in the layer impedance matrix have the same dimensions than the surface impedance matrices.

Although the computation of layer impedance matrix for many layers is not used in the surface impedance matrix method, it is recalled here for comparison with the iterative computation of the surface impedance matrices. The impedance matrix $[Z^{1\cdots j+1}]$ of the $j+1$ first layers, can be computed from the impedance matrix of the j first layers $[Z^{1\cdots j}]$ and the impedance matrix $[Z^{j+1}]$ of layer $j+1$ and by eliminating $[\boldsymbol{\sigma}^j]$:

$$[Z^{1\cdots j+1}] = \begin{bmatrix} [Z_{11}^{1\cdots j}] + [Z_{12}^{1\cdots j}][z][Z_{21}^{1\cdots j}] & -[Z_{12}^{1\cdots j}][z][Z_{12}^{j+1}] \\ [Z_{21}^{j+1}][z][Z_{21}^{1\cdots j}] & [Z_{22}^{j+1}] - [Z_{21}^{j+1}][z][Z_{12}^{j+1}] \end{bmatrix}, \tag{3}$$

where $[z] = ([Z_{11}^{j+1}] - [Z_{22}^{1\cdots j}])^{-1}$ and $[Z_{pq}^{1\cdots j}]$ $(p,q=1,2)$ are the sub-matrices of the j first layers. As described by Wang and Rokhlin [4–8], Eq. (3) defines the algorithm for computing the total impedance matrix of the stratified plate.

4. Surface impedance matrices

Media 0 and $n+1$ (Fig. 1) are supposed to be semi-infinite spaces made of fluid, solid or vacuum. There is no wave going down in layer 0 and no wave going up in layer $n+1$. At most three partial waves are going up in medium 0 and down in medium $n+1$. Only the sub-matrices $[Z_{22}^0]$ and $[Z_{11}^{n+1}]$ are not null and the surface impedance matrices $[ZI_u^0] = [Z_{22}^0]$, $[ZI_d^n] = [Z_{11}^{n+1}]$ link stresses to particle velocities at interfaces 0 and n respectively. They are the impedances of the top ("Looking up") and bottom ("Looking down") semi-infinite media, respectively. In case of vacuum, they are null: $[ZI_u^0] = 0$ and/or $[ZI_d^n] = 0$. For a fluid, only one component $[ZI_{11}]$ is not null. This constitutes the starting point to compute iteratively all the surface impedance matrices.

"Going down" and "Looking up", at interface $j-1$, in the layer $j-1$, the surface impedance matrix is

defined by $[\boldsymbol{\sigma}^{j-1}] = [ZI_u^{j-1}][\mathbf{V}^{j-1}]$ (Fig. 1). In the layer j, between interfaces $j-1$ and j, the layer impedance matrix leads to the relations:

$$[\mathbf{V}^{j-1}] = -[\widetilde{Z}_{11}^{j}]^{-1}[Z_{12}^{j}][\mathbf{V}^{j}] \quad \text{and}$$

$$[\boldsymbol{\sigma}^{j}] = -([Z_{21}^{j}][\widetilde{Z}_{11}^{j}]^{-1}[Z_{12}^{j}] + [Z_{22}^{j}])[\mathbf{V}^{j}].$$

This defines the surface impedance matrix at interface j as a function of the surface impedance matrix at interface $j-1$ and the impedance matrix in the layer j:

$$[ZI_u^{j}] = [Z_{22}^{j}] - [Z_{21}^{j}][\widetilde{Z}_{11}^{j}]^{-1}[Z_{12}^{j}], \quad \text{with}$$

$$[\widetilde{Z}_{11}^{j}] = [Z_{11}^{j}] - [ZI_u^{j-1}], \tag{4}$$

and reciprocally, "Going up" and "Looking down":

$$[ZI_d^{j-1}] = [Z_{11}^{j}] - [Z_{12}^{j}][\widetilde{Z}_{22}^{j}]^{-1}[Z_{21}^{j}], \quad \text{with}$$

$$[\widetilde{Z}_{22}^{j}] = [Z_{22}^{j}] - [ZI_d^{j}]. \tag{5}$$

These equations (4) and (5) are fully similar to the recursive equation introduced by Rokhlin and Wang in Ref. [7] to get the characteristic equation for surface waves in a semi-infinite layered structure.

5. Modal solutions

5.1. Dispersion equations

"Going down", starting with the impedance matrix at interface 0, $[ZI_u^0]$, relation (4) supplies the impedance at the last interface of the medium above that interface ("Looking up") $[ZI_u^n]$. According to the continuity of the stresses at interface n: $([ZI_u^n] - [ZI_d^n])[V^n] = 0$. Then the modal solutions are deduced from the dispersion equation: $|[ZI_u^n] - [ZI_d^n]| = 0$.

Reciprocally, the impedance matrix $[ZI_d^0]$ at interface 0 of the medium below that interface ("Looking down") is computed successively from that at the last one, "Going up", and the continuity of the stresses at interface 0 leads to the dispersion equation: $|[ZI_u^0] - [ZI_d^0]| = 0$.

More generally, the modal solutions can be deduced from the dispersion relation at any interface j:

$$|[\Delta ZI^{j}]| = 0, \quad \text{with}$$

$$[\Delta ZI^{j}] = [ZI_u^{j}] - [ZI_d^{j}] \quad \text{and} \quad 0 \leqslant j \leqslant n, \tag{6}$$

where the surface impedance matrices $[ZI_d^j]$ and $[ZI_u^j]$ are computed "Going up" from interface n to interface j and "Going down" from interface 0 to interface j, respectively. Eq. (6) is a generalization to a multilayer stratified medium of equation: $\det(\vec{Z} + \overleftarrow{Z}) = 0$, given in Ref. [9]. It follows from Eq. (21) of Ref. [6], where the modal solutions are the roots of the Green's function obtained in Ref. [6].

To find the modal solutions for modes concentrated in only one or in a few layers, it is judicious to select an interface belonging to one of these layers, since the particle velocities and stresses can be null elsewhere. The roots of Eq. (6) form the dispersion curves that represent the complex wave numbers of modes propagating along the structure versus frequency.

5.2. Speed computation and stability

One interest of the surface impedance matrix method is to reduce the number of operations. To connect one layer to the following, the layer impedance matrix method [4–8] implies six multiplications of sub-matrices in Eq. (3), while the surface impedance matrix method in Eqs. (4) or (5), uses 2 multiplications only for linking one interface to the other.

Furthermore, to find the modal solutions, the dispersion equation (6) obtained from the surface impedance matrix method leads to a matrix that is four times smaller than that using the layer impedance matrix method. Typically, coding the two methods in the same language, on the same machine and running them with the same input data, lead to at least a factor of 2 between the total computation times.

As mentioned in Refs. [4–8], the layer impedance matrix method is stable since it involves only the inversion of matrices $[Z_{11}]$ and $[Z_{22}]$ and these matrices are not singular when the frequency × thickness of a layer is large, except for the particular case where $[Z_{11}^{j+1}] = [Z_{22}^{1\ldots j}]$ so that $[z]$ is singular. For the same reason, the iterative algorithm for computing the surface impedance matrices is generally stable.

However, Eqs. (4) or (5) involve the inversion of sub-matrices $[\widetilde{Z}_{11}^{j}]$ or $[\widetilde{Z}_{22}^{j}]$ and leads to instabilities if $[Z_{11}^{j}] = [ZI_u^{j-1}]$ or $[Z_{22}^{j}] = [ZI_d^{j}]$. These cases correspond to null particle velocities at interface $j-1$ or j respectively. These numerical instabilities can be easily avoided by choosing another and more suitable interface to satisfy the boundary conditions and to solve Eq. (6).

6. Particle velocity and stress fields computation inside the plate

6.1. Particle velocity link between interfaces

From any modal solution, all the surface impedance matrices $[ZI_d^m]$ can be numerically computed "Going up", while the particle velocity field is computed "Going down". If the particle velocity field is known at interface $j-1$, the particle velocity field at interface j is:

$$[V^{j}] = -[\widetilde{Z}_{22}^{j}]^{-1}[Z_{21}^{j}][V^{j-1}], \tag{7}$$

and reciprocally:

$$[V^{j-1}] = -[\widetilde{Z}_{11}^{j}]^{-1}[Z_{12}^{j}][V^{j}]. \tag{8}$$

The stresses at any interface will be deduced from the impedance matrices of this interface using Eq. (1).

6.2. Initial displacement

To find the total particle velocity field in the structure, the particle velocity field at one specific interface must be initially set. For a given mode, at interface j, the particle velocity field must satisfy the relation:

$$[\Delta ZI^j][V^j] = 0. \tag{9}$$

The component of the particle velocity normal to the interface is taken arbitrarily equal to 1 ($v_1^j = 1$) and the other unknown components of $[V^j]$ are deduced from the relation (9). In the case of 2D, the second component of the particle velocity is computed from: $v_2^j = -\Delta Z_{21}^j / \Delta Z_{22}^j$. For some modes, at interface j, v_1^j can be null and in consequence ΔZ_{22}^j is also null. This causes numerical instabilities when computing the through thickness particle velocity and displacement fields. Nevertheless it is easy to check if ΔZ_{22}^j is smaller than ΔZ_{11}^j and, in that case, to impose $v_2^j = 1$ and then calculate $v_1^j = -\Delta Z_{12}^j / \Delta Z_{11}^j$. The other alternative when both terms ΔZ_{11}^j and ΔZ_{22}^j have very small values, is to choose another interface as a starting point for the through-thickness fields computation.

One of the advantages of the surface impedance matrix method is the possibility to choose easily the initial interface to execute the computation of the particle velocity field. This is crucial for channel modes.

Subsequently the particle velocities, amplitudes and stresses inside any layer j are computed in a similar way that with the layer stiffness method [4–8].

6.3. Surface impedance matrices at solid/fluid interfaces

In a fluid layer, only one longitudinal mode exists. The surface impedance matrix method is well suited to include fluid layers between solid layers. The layers j and $j+1$ are supposed to be fluid and solid respectively. The computation "Going up" gives the surface $[ZI_d^j]$ that is a 2×2 matrix in the solid, at the interface j separating the two layers. The normal stress is continuous and the presence of the fluid imposes the shear stresses to be null at this interface. Therefore, when the computation "Going up" is developed, the surface impedance matrix becomes a scalar at the solid/fluid interface.

When crossing a fluid to solid interface, the surface impedance matrix becomes a scalar. Then, the presence of a fluid layer in the stratified medium reduces again the number of computations.

7. Example: dispersion curves for channel modes

In simple situations, for instance for a limited number of layers and symmetry with respect to the middle plane, closed form solutions can be found [12,13]. The surface impedance matrix method is tested for such a case to check its ability to find modes that propagate along a thin fluid layer embedded between two thicker solid layers.

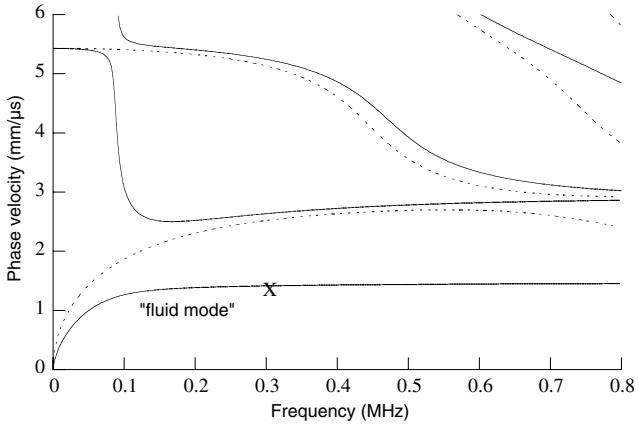


Fig. 2. Dispersion curves for aluminium/water/aluminium structure. Solid line: symmetric modes. Dashed line: antisymmetric modes.

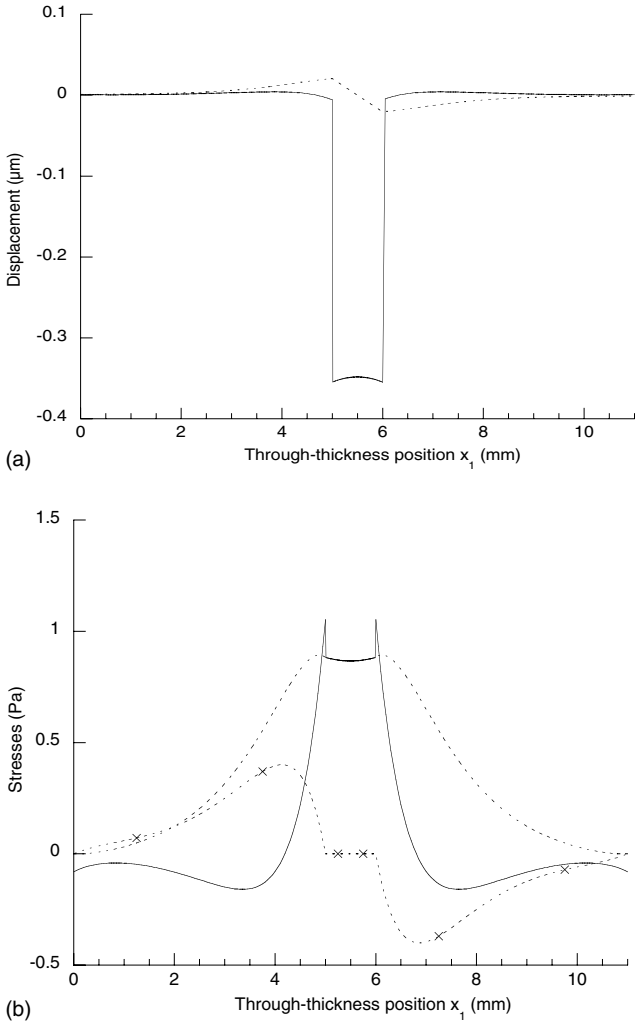


Fig. 3. Power-normalized (a) displacements and (b) stresses in the aluminium/water/aluminium structure for the "fluid" mode, at 0.3 MHz. (—): U_2, σ_2, (- - -): U_1, σ_1, and (-×-): σ_6.

A free layered plate composed of two aluminium layers (5 mm thick) with a water layer (1 mm thick) in between was investigated theoretically and experimentally [12,13]. A mode called "fluid mode" that carries energy essentially in the water layer, was found. The surface impedance matrix method, applied to interface $n = 0$ or 3, finds all the modes except the "fluid mode" above around 0.3 MHz (point noted X in Fig. 2). When considering the interface 1 (fluid–solid) or 2 (solid–fluid), $j = 1$ or 2 in Eq. (6), all the modes are found. That can be explained by Fig. 3 that presents the displacements and stresses inside the structure, normalized by the power flow carried by the mode through the whole plate thickness and over a temporal period. Fig. 4 shows the variation of the displacement component u_1 normal to the interfaces for 3 frequencies. To obtain these results, the computations were done using interface 1 as the initial interface. When using the first ($j = 0$) or the last interface ($j = 3$) as the starting interface to compute the displacement field, the results were found erroneous and unstable above 0.4 MHz. The amplitudes of the displacements and stresses in the solid layers are decreasing as the frequency is increasing, for the "fluid mode". The energy of this mode is mostly contained in the fluid layer, above 0.3 MHz. However, the name

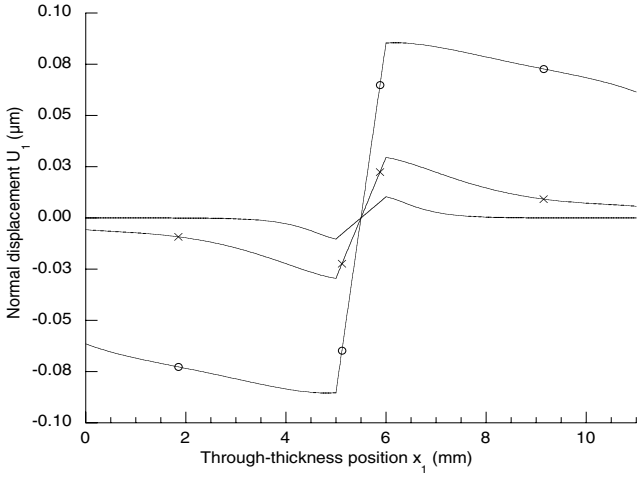


Fig. 4. Power-normalized displacement U_1 in the aluminium/water/aluminium structure for the "fluid" mode, at three frequencies: (–O–): 0.1 MHz, (–×–): 0.2MHz, (—): 0.6 MHz.

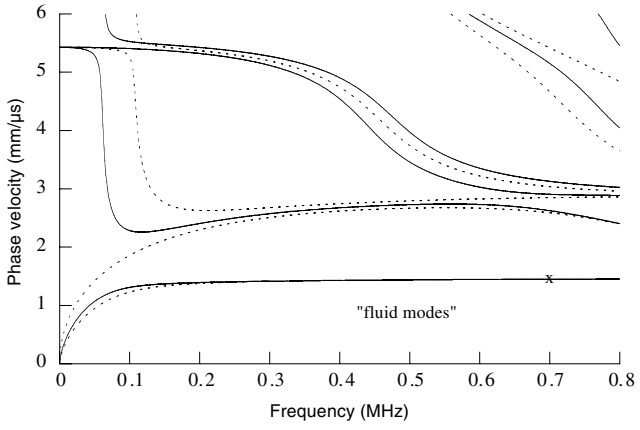


Fig. 5. Dispersion curves for aluminium/water/aluminium/water/aluminium structure Solid line: symmetric modes. Dashed line: antisymmetric modes.

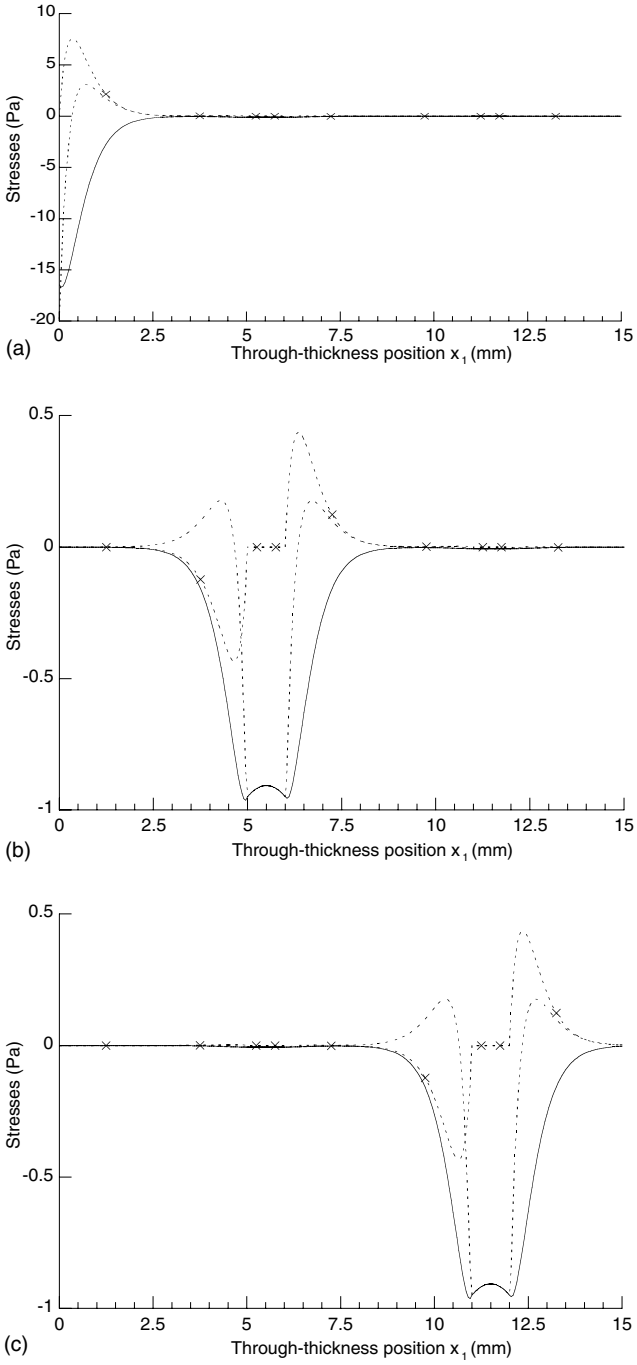


Fig. 6. Power-normalized stresses in the aluminium/water/aluminium structure for the "fluid" modes, at 0.7 MHz. (a) Interface 0, (b) interface 2 and (c) interface 3 (—): σ_2, (- - -): σ_1, and (-×-): σ_6.

"fluid mode" is appropriate only at high frequency. For low frequency, it was found that the mode also carries energy in the aluminium layers.

Another more complex structure, made of the same layers with the stacking sequence of aluminum/water/aluminum/water/aluminum is now investigated for clarifying the rule of the interface choice. Fig. 5 shows the new modes due to the addition of two more layers and that the "fluid modes" have the same velocity that the bulk mode phase velocity in water. To conclude that there are two "fluid modes" propagating at the same velocity, one must choose various interfaces to compute the initial field and consequently the field in the whole structure. Fig. 6 presents the power-normalized stresses in the five layers structure for the "fluid" modes, at 0.7 MHz. When computed from interface 0 (Fig. 6a), the solution is wrong since the normal stress is not null at the surface. This is due to the fact that the displacements are null at this interface and one should not start the computation from here. When the computation starts from interface 2 (Fig. 6b) or interface 4 (Fig. 6c), the stress fields correspond to a channel mode propagating in the first water layer and in the second layer respectively.

This example shows the interest of the surface impedance matrix method to investigate the propagation of guided modes in a multilayered structure at any interface or in any layer.

8. Conclusions

The surface impedance matrices that connects the stress field to the particle velocity field at any interface in a stratified medium have been introduced in iterative methods for modeling the propagation of plane waves in stratified plates.

In comparison with the transfer matrix methods, it makes divided by a factor of four the number of matrix elements, thus reducing considerably the number of operations and therefore the computation times.

The surface impedance matrix method eliminates numerical instabilities of the transfer matrix methods, in the case of channel modes propagating along embedded layers. Such waves possess non-zero displacements and stresses in a limited number of layers only inside the medium. The surface impedance matrix method offers the possibility to select a specific interface where significant displacements exist, thus making stable computations of the modal solutions and of the particle velocity and stress fields through the whole structure.

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